DOUBLE JEOPARDY

Do Coastal Rules Reach the People at Risk? Comparing Five Island Jurisdictions

Executive Summary · Version 2.1 · DOI: 10.5281/zenodo.23827474 · Sakshi D. Maske

Project Overview

Low islands face a rising sea while the reefs and mangroves that shelter them are under stress. This project measures both across 5 island groups in 3 ocean basins and compares how their governments regulate the coast. It is a study in **comparative environmental policy**: the five cases sit under five different governments, from three sovereign small island states (Maldives, Seychelles, Fiji) to an Indian Union Territory whose coastal rules come from New Delhi (Lakshadweep) and an outermost region of the European Union (the Canary Islands), which makes **multi-level governance** part of the comparison.

Version 1 reported Seychelles as most exposed, significant coral trends, a ranking that held at every weighting and protection that tracked risk. Rechecking each measurement choice withdrew most of those results. Version 2 shows how much the answer depends on choices that usually go unreported, such as which height data is used or whether people or places are counted.

The Question

Are the governments' coastal rules drawn where the people and buildings at risk are, and do counts of coastal instruments say anything about that fit? Behind that sit three measurement questions. How many people on these islands live at or below 1 m? How has heat stress on their reefs changed between 1996 and 2020, and have mangroves declined? And does protected-area coverage follow the combined risk?

The Method

- **Policy instruments:** each government's coastal instruments classified with **Hood's NATO tools**. The **Authority** instrument tested is the statutory building setback: 100 m in the Canary Islands (Ley de Costas), 25 m in Seychelles (S.I. 91 of 2023), 20 m in Lakshadweep (ICRZ 2019), and none for inhabited land in the Maldives and Fiji.
- **Research design:** a structured comparison of five jurisdictions facing the same hazard. The Maldives and Lakshadweep are the closest pair to a **most similar** comparison, the same kind of coral atoll under a sovereign state and under India's national coastal regulation, though they also differ in population, reclamation and tourism.
- **Policy outputs and design fit:** coastal rules in force in October 2026 coded by hand from the legal texts, as **policy density** (six instrument types) and **policy intensity** (setback width), following Knill, Schulze and Tosun (2012). Every coded value carries its legal source. Separately, at **building level**, about 1.35 million building footprints (Overture Maps, 2026-08-19.0), each with its bare-earth elevation and distance to the shoreline, show whether a setback's width matches where low buildings stand (**policy design**), not whether the rules are complied with.
- **Policy change:** 23 dated rule changes since 1988, coded as expansion, **policy dismantling** or a shift between levels of government, with the V-Dem electoral democracy index for each year of change.
- **Exposure:** share of people (WorldPop 2020, 100 m) on land at or below 1 m on the **DeltaDTM v1.1 bare-earth model** (Pronk et al. 2024), with the Copernicus GLO-30 surface model and the share of

OpenStreetMap named places for comparison. Heights are above the EGM2008 geoid.

- **Reef heat stress and mangroves:** NOAA Coral Reef Watch Degree Heating Weeks averaged over mapped reef, 1996–2020, with a seasonal Mann-Kendall test corrected for serial correlation; Global Mangrove Watch v3 at 1996, 2010 and 2020.
- **Compound score and protection:** exposure and change in reef heat stress, each scaled 0–1, with the weighting swept from 0 to 100%; share of the 10 km coastal zone inside a protected area (WDPA, June 2026), described only.

The Findings

Coastal setbacks against low-lying buildings

Island group	Buildings at or below 1 m	Statutory setback	Low buildings within the setback	Within 20 m, same width everywhere	Distance covering 90% of them
Maldives	13.4% (15,392)	none for inhabited islands	no setback	5.2%	225 m
Lakshadweep	4.7% (444)	20 m	6.8%	6.8%	345 m
Seychelles	4.6% (2,440)	25 m	8.0%	5.6%	427 m
Fiji	0.5% (1,397)	none	no setback	6.9%	463 m
Canary Islands	0.02% (192)	100 m (20 m urban)	84.9%	51.6%	122 m

In Lakshadweep and Seychelles the setbacks cover 7–8% of the low buildings, which on the four tropical island groups stand a median 80–136 m from the shore. At the same 20 m width, the Maldives and Fiji, with no rule, would cover 5–7%, so coverage follows island geometry rather than the rule. In the Canary Islands the 100 m line covers 85% of 192 low buildings, a thin beachfront and port strip (52% under the 20 m rule for urban land). Measured on low land, which is what a setback governs for new building, a 20–25 m setback covers 19–26% of the ground at or below 1 m on the atolls and Seychelles, and a 100 m line 63–70%. On the Maldives even a 100 m rule would cover only 54% of the low buildings and would put 39% of all buildings inside it, because the low ground on an atoll reaches far inland.

Policy outputs against design fit

	Canary Islands	Lakshadweep	Seychelles	Fiji	Maldives
Coastal instrument types in force (of 6)	6	5	3	3	2
Statutory setback	100 m (20 m urban)	20 m	25 m	none	none
Low buildings within 20 m of the shore, same width	51.6%	6.8%	5.6%	6.9%	5.2%

Instrument counts and coverage do not line up. Lakshadweep has five of six instrument types and the Maldives two, yet a distance rule fits the low buildings of both equally poorly. Neither measure shows **policy performance**, which would need evidence on compliance and implementation (Daugbjerg 2022). Of 23 rule changes since 1988, 15 added protection. Spain relaxed its coastal law in 2013 and the Maldives widened exemptions from environmental review in 2024–2025, and in 2023 Spain handed part of coastal management to the Canary Islands government. Regime type does not sort these changes, since rules were added under V-Dem scores from 0.13 to 0.89 and relaxed at 0.88 (Spain) and 0.56–0.58 (Maldives).

People at or below 1 m

Island group	Bare earth (DeltaDTM)	Surface (Copernicus)	People
Maldives	11.82%	8.88%	57,889
Lakshadweep	6.40%	0.79%	4,238
Seychelles	2.07%	0.43%	1,933
Fiji	0.28%	0.21%	2,460
Canary Islands	0.19%	0.15%	3,946

Hypothesis	Result
H1: Reef heat stress is rising	**Partly supported.** Mean DHW over reef rose on every reef island (Seychelles +0.52, Fiji +0.29, Maldives +0.07, Lakshadweep +0.05 °C-weeks), but no monotonic trend is significant once serial correlation is accounted for ($p = 0.14$ to 0.90)
H2: Mangroves are declining	**Not supported.** No decline on any of the 3 islands (Fiji 485.7 → 488.4 km^2)
H3: Protection follows risk	**Not supported.** $r = -0.10$ (95% interval −0.97 to 0.95, $n = 4$); coverage from 1.3% (Maldives) to 71.4% (Seychelles)

Compound score (equal weights): Seychelles 0.58 · Maldives 0.56 · Lakshadweep 0.32 · Fiji 0.28 · Canary Islands 0.00. Exposure is highest in the Maldives and the rise in reef heat stress largest in Seychelles, so the Maldives move to the top once exposure carries 52% or more of the weight. At equal weights the two are almost level.

What Changed From Version 1, and Why

- **Withdrawn: "78.3% of Seychelles settlements at risk."** 188 of 244 Seychelles points read the elevation raster's NoData value (−32767 m) and were counted as at risk, and most OSM "settlement" points on these islands are names of islands and islets, some in open water. I now count people, with named places as a second measure (Seychelles 24.7%, Maldives 28.8% on bare earth).
- **Withdrawn: surface elevation model only.** Copernicus measures roofs and canopy on wooded atolls. On bare earth, Lakshadweep goes from 0.8% to 6.4% of people at or below 1 m. Where Copernicus is above 30 m, I do not use DeltaDTM, since on small forested granite islands in Seychelles it reads a hill as sea level.
- **Withdrawn: "Coral trends significant for Maldives ($p = 0.0046$) and Seychelles ($p = 0.0069$)."** That used one 5 km cell per island and a test that treats neighbouring months as independent, which they are not for a 12-week running sum. With the correction, Seychelles $p = 0.14$ and Maldives $p = 0.31$.

- **Withdrawn: "The ranking holds at every weighting."** Seychelles came out top at every weighting because it was highest on both inputs by construction. With people and reef-cell means, the inputs disagree and the ranking depends on the weights.
- **Withdrawn: "Governance correlation r = 0.965, p = 0.035."** Overlapping protected areas were counted twice and coverage was divided by land area. Measured on the coastal zone, the relationship disappears.
- **Withdrawn: Seychelles built-up rise (+0.11 NDBI).** It was measured over bounding boxes that were mostly ocean. On land only, Seychelles shows no rise and the Maldives a small one (+0.055).

Limitations

- Exposure is an elevation threshold with no tides, surge, waves or connectivity to the sea, and DeltaDTM's error against airborne lidar (about 0.4–0.5 m) is large next to 1 m. The comparison between models is more reliable than any single percentage.
- WorldPop is a modelled population surface and can differ from census counts.
- Five island groups (four for governance) describe this sample and cannot establish general relationships.
- With min–max scaling, the Canary Islands' 0.00 is the lowest score in this sample and does not mean zero risk. The score also leaves out adaptive capacity.
- Mangrove patches on the atolls are close to Global Mangrove Watch's pixel size, so small losses may go undetected.
- Setback coverage measures design fit on today's building stock, much of which predates the rules, so it is not evidence of compliance. The Canary figure rests on 192 buildings and a flat 100 m width.
- Setback distances run to the island outline instead of the legal high water mark, and only national law is compared; local plans, permits and enforcement are not observed. The Lakshadweep width and the Maldivian changes of 2024–2025 rest on secondary sources, and the policy coding was done by one coder.
- Independent recalculations of the coral trend test and of the setback distances (on the WGS84 ellipsoid) reproduce the reported results.

Policy Relevance

For coastal law on atolls, the results point to rules written in heights, such as minimum floor or ground levels, since a setback measured from the shore misses most of the buildings exposed to **coastal flood risk**. For comparative policy research, they show why counts of policy outputs should sit next to a check of whether the instruments fit the hazard: counting instruments would rank Lakshadweep far ahead of the Maldives, yet a distance rule fits both equally poorly. Adaptation finance for small islands is also increasingly tied to vulnerability indices, and the same islands can come out in a different order just by changing the height data or the weights, so any such index should state its choices. A two-page policy brief sets this out for practitioners (`paper/DJ_Policy_Brief.md`).

GitHub: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY | Live Dashboard: double-jeopardy-6ev9trz3dwafsb7panbnxg.streamlit.app | Zenodo DOI: 10.5281/zenodo.23827474

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